

MC8P MIDI Controller - Build Guide

.axs instruments

Bill of Materials

MC8P B.O.M	
Component	Qty
Teensy 4.0	1
Kailh Mechanical Switch	4
B100k Potentiometer	8
10k Resistor	4
220Ω Resistor	2
10nF Capacitor	8
128x64 SSD1306 OLED Monochrome Display	1
PJ307 3.5mm Headphone Socket	1
M3 10mm (Female-Female) Stand-offs	4
M3 10mm (Female-Male) Stand-offs	4
M3 Screws	4
M3 Nuts	4
M2 5mm (Female-Male) Stand-offs	2
M2 Screws	2
M2 Nuts	2

Build Guide

Smaller components

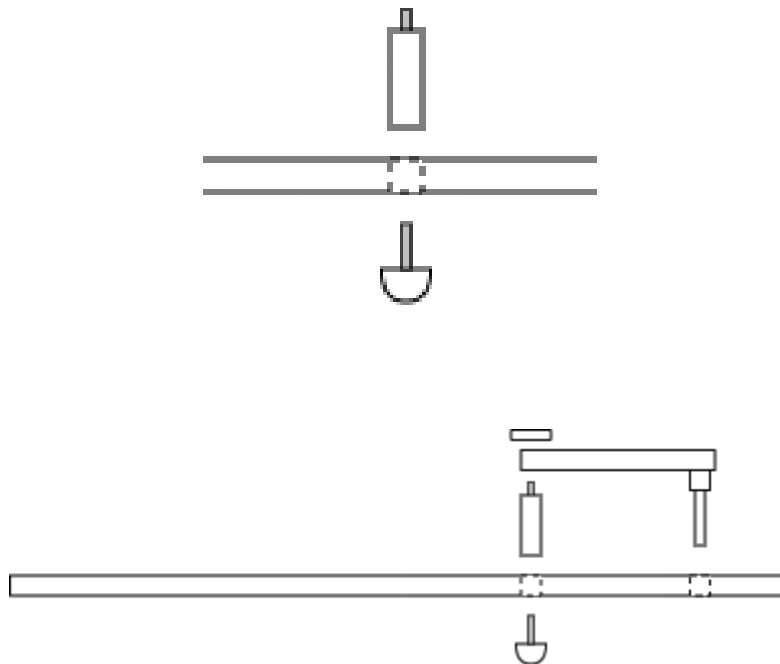
- Solder resistors to the **Main Board**: R1–R4 (10k pull-down), R5 & R6 (MIDI Out).
- Solder ceramic capacitors: C1–C8.
- Install PJ307 headphone socket (J1).

Teensy 4.0

- Solder header pins to Teensy.
- Mount and solder Teensy to the board.

OLED Display (SSD1306)

- Flip board, insert 2× M2 screws through bottom holes.
- Attach female-to-male M2 stand-offs from front.
- Solder pin headers to the back of OLED.
- Align OLED with stand-offs and header holes.
- Secure with M2 nuts.
- Solder pin headers to **Main Board**



Switches & Potentiometers

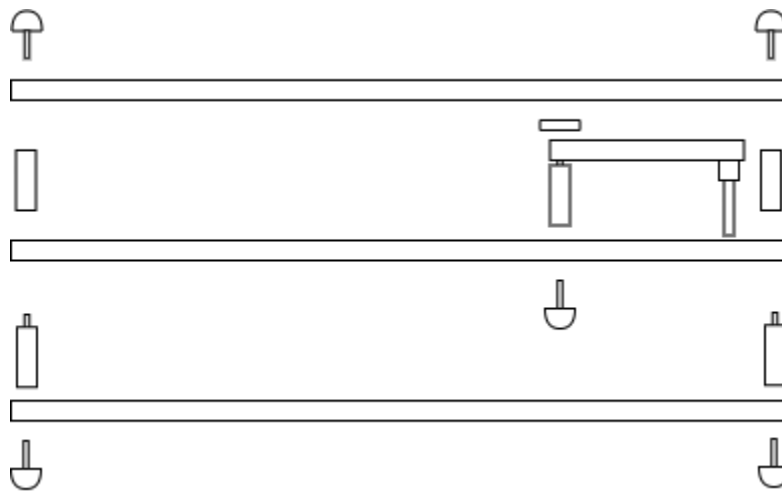
- Solder Kailh switches (SW1–SW4) to the **Main Board**.
- Solder potentiometers (RV1–RV8) to the Main Board.

Back Panel Assembly

- Insert 4× M2 screws (with nuts screwed on) through back panel holes.
- Attach 4× 10mm M3 female-male stand-offs to the blank side.
- Mount main board and secure with M3 female-female stand-offs.

Front Panel & Final Assembly

- Attach the front panel with remaining M3 screws + nuts.
- Secure potentiometers with washers and nuts.
- Add potentiometer knobs.



That's it!